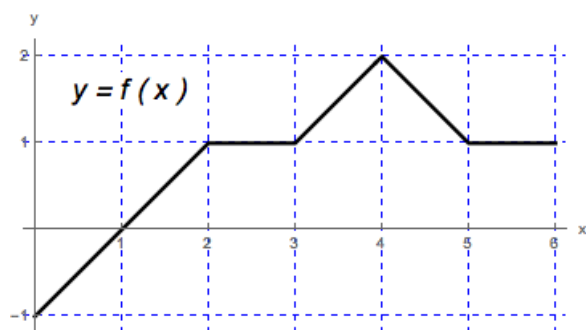


Problem 5

Problem 5

Problem 1 The graph of a function f defined on the interval $[0, 6]$ is given in the figure.



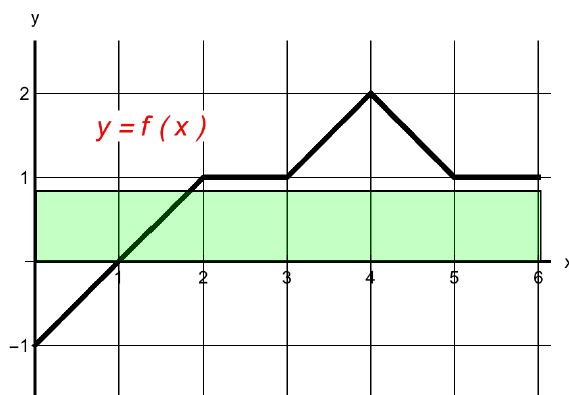
- (a) Compute the net area of the region between the graph of f and the x -axis, on the interval $[0, 6]$. [3]

Explanation. The net area $= -1/2 + 1/2 + 1 + 3 + 1 = 5$

- (b) Draw a rectangle with base on the x -axis, $0 \leq x \leq 6$, whose area is equal to the net area in part (a).

Explanation. Let A be the area of the rectangle. Then $A = (\text{base})(\text{height}) = 6h = 5$.

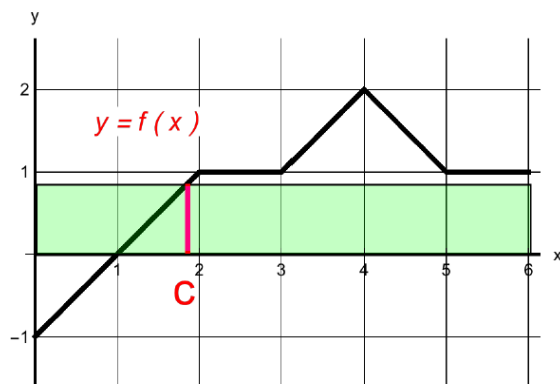
So, $6h = 5 \implies h = \frac{5}{6}$



Problem 5

- (c) In the figure, mark a point c in $(0, 6)$ such that $f(c)$ is the height of the rectangle from part (b).

Explanation. $f(c) = h = \frac{5}{6}$



Note: In this example, there is only one such point c . In some cases there may be more than one.

- (d) Using the figure and parts (a-c), what is the relationship between the rectangle from part (b), the net area from part (a), and the average value of f on $[0, 6]$? [3]

Explanation. The Mean Value Theorem for integrals states that there exists a point d in (a, b) such that

$$f(d) = \frac{1}{b-a} \int_a^b f(x) dx.$$

The right-hand side of this equation is the average value of f on $[a, b]$.

Rewriting we have

$$f(d) \cdot (b-a) = \int_a^b f(x) dx.$$

The right-hand side of this equation is the net area found in part (a):

$$\int_0^6 f(x) dx = 5.$$

The left-hand side of this equation is the area of the rectangle in part (b).

The point d is the point c we found in part (c). From (b) and (c), $f(c) = \frac{5}{6}$ is the average value of f on $[0, 6]$.